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Properties of Materials
Lecture 7

Dr. Alkis Tsamis
at441@leicester.ac.uk

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Lecture 4: Availability of Materials, Case Study, Exam Question 2013, Example Questions, The World of Materials (2), Materials Selection in Design (2), Properties of Engineering Materials (3), Three-Way Relationship (2), Learning Objectives, Why Do We Need a Selection Methodology, Mechanical Properties (3), Thermal Properties (3), Trade-Offs Must Be Made, Materials Selection in Design (3), Materials Selection Process (1)
Lecture 5: Step 1 - Translation, Step 2 - Screening, Materials Selection Process (2), Step 3 - Ranking Materials, Example - Hydraulic Cylinder Tie Rods, Step 4 - Refining Selection, Concepts and Terminology (2)
Lecture 6: The Elastic Moduli, Deformation of Brittle Materials, Deformation of Ductile Materials, Hooke's Law - Young's Modulus, Linear Elastic Behaviour, Structural Engineering, What Might Have Saved the Bridge, Engineering Stress and Strain, True Stress and Strain, Comparison
Lecture 7: Practice Time, Definitions, Concepts and Terminology (3), Task

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The elastic moduli

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Practice time!

- Draw a graph of nominal stress vs nominal strain for a typical ductile metal
- Label the following:
  - **Failure stress** and **strain to failure**
  - **Yield stress**
  - **Tensile strength**
  - **Elastic** and **plastic** regions
- Define the terms **elastic**, **plastic** and **yield**

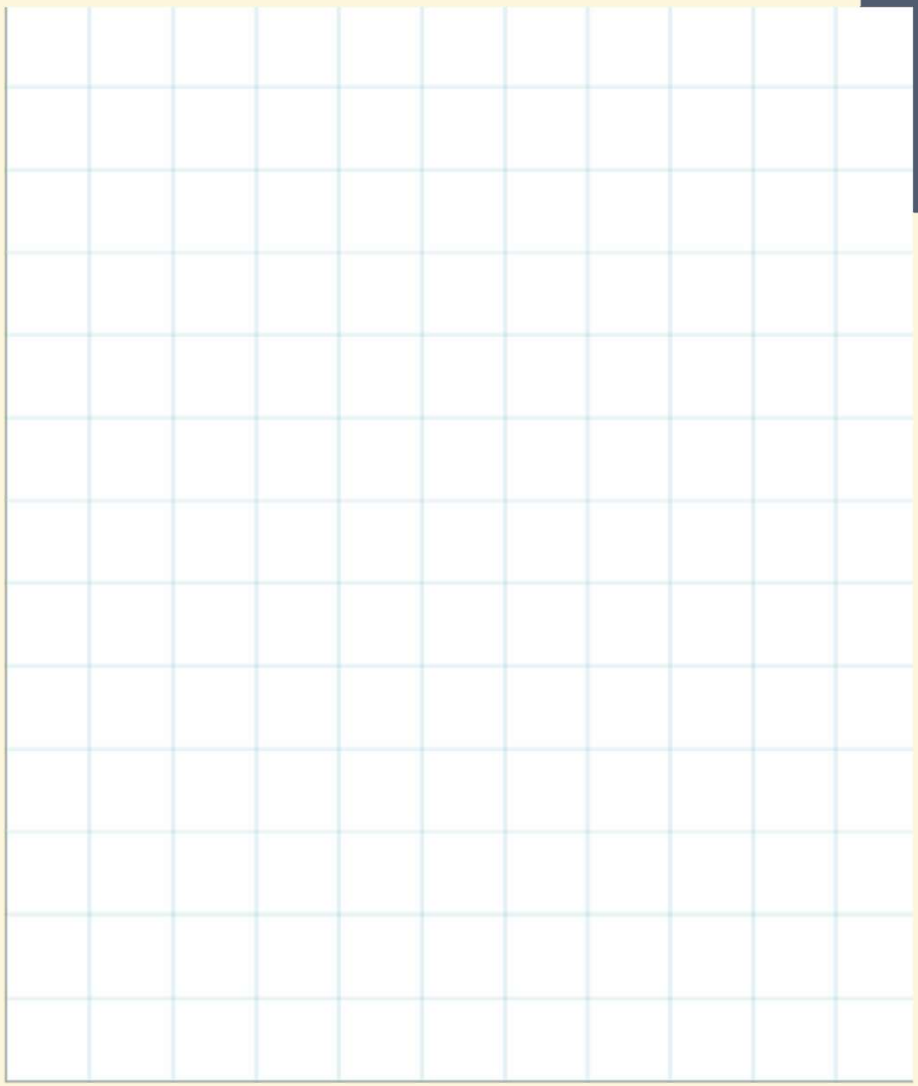
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- Why is yield stress rather than tensile strength typically used in design?

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Nominal stress vs nominal strain for a ductile metal

- Draw a graph of **nominal stress vs nominal strain** for a typical **ductile** metal
- Label the following:
  - **Failure stress** and **strain to failure**
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- Why is yield stress rather than tensile strength typically used in design?



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Definitions

- Elastic deformation is when…
  - A material is deformed **proportional to an applied stress**, but returns to its original shape when the stress is removed
- Plastic deformation is when…
  - A material is **permanently deformed** due to an applied stress
- Yield:
  - The **point at which a material ceases to behave elastically**, beyond which it will behave plastically

Summary

- Brittle materials have a single, **linear elastic** region to failure
- Ductile materials undergo elastic deformation before **yielding** followed by **plastic deformation** before failure
- We can model the stiffness of materials by a 'ball and spring' model of atomic attraction
- Engineering or Nominal stress/strain is based on the **initial dimensions** of a test piece
- True stress/strain is based on the **instantaneous dimensions** of the test piece at any point in time

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Concepts and terminology

- Stress-strain curves:
  - **Brittle** and **ductile** materials
- **Stiffness**:
  - Hooke's Law
  - Young's modulus
- **Linear elasticity**
- Engineering / **nominal** stress and strain
- **True** stress and strain

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Recap from last time…

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Hooke's Law, Young's Modulus…

…and the longest derivation in history!

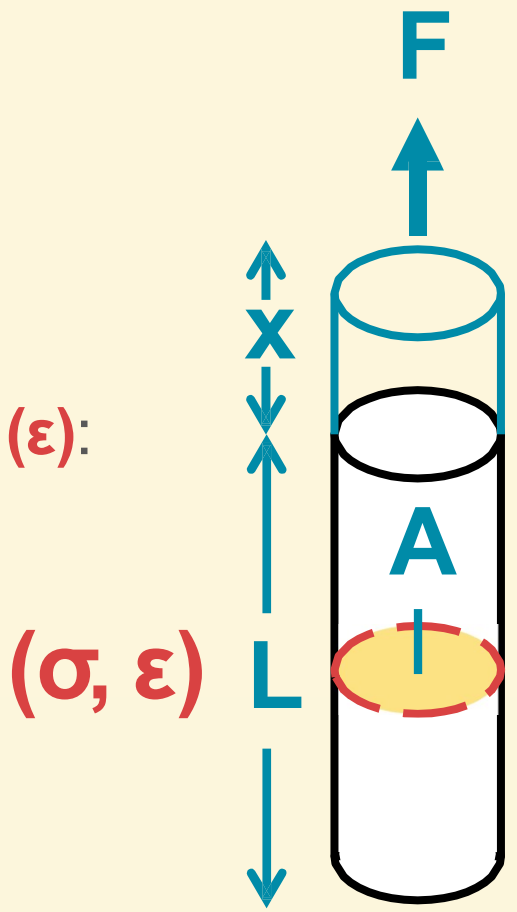
- Hooke's Law (1678): *"ut tensio, sic vis" - Hooke, 1660*
  - Force (F) = Stiffness (k) × Extension (x)

- Divide through by Area (A) to develop **Stress (σ)**:
  - $\sigma = \frac{kx}{A}$

- Multiply and divide by Length (L) to develop **Strain (ε)**:

$$-\sigma = \frac{kL}{A}\varepsilon \Rightarrow \frac{kL}{A} = \frac{\sigma}{\varepsilon}$$

- Young's Modulus (1807):
  - Elastic Modulus (E) = $\frac{\text{Stress}(\sigma)}{\text{Strain}(\varepsilon)}$



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Linear elastic behaviour

- Ball and spring model of elasticity:
- Material in tension –bonds stretched and atoms move apart
- Material in compression –bonds compressed and atoms move closer together

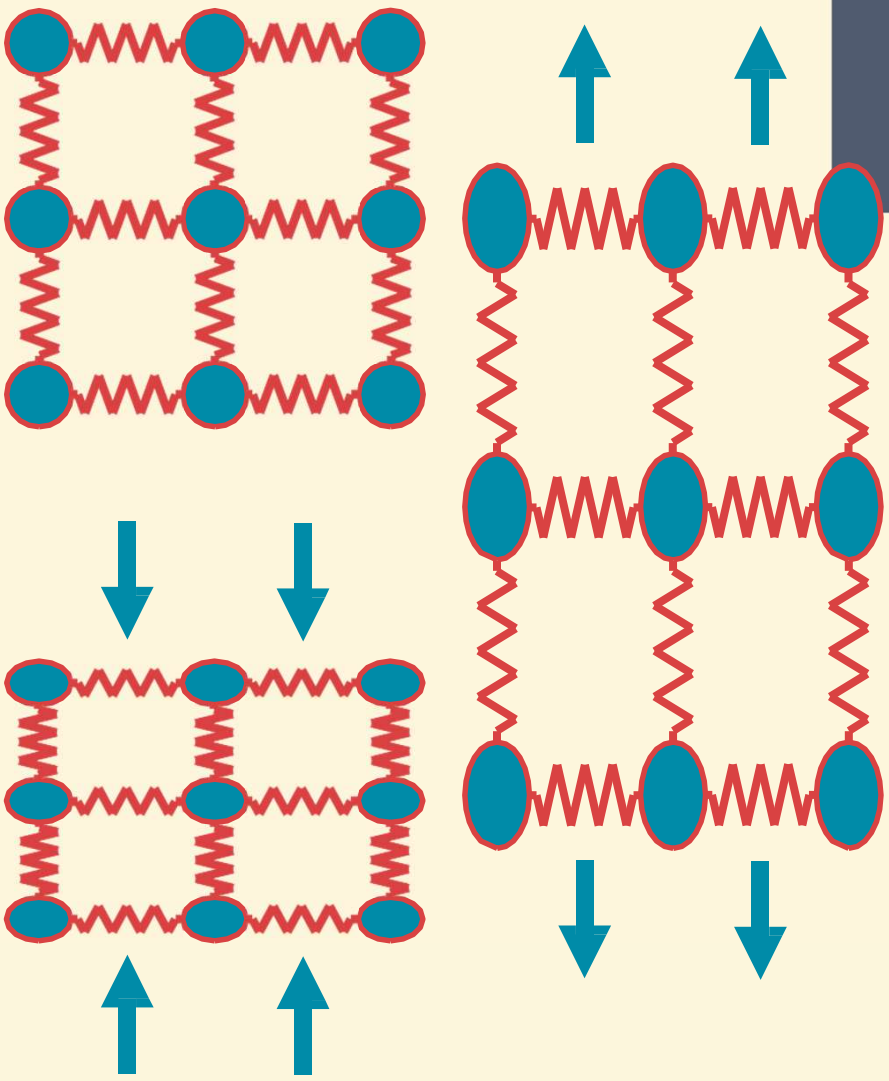
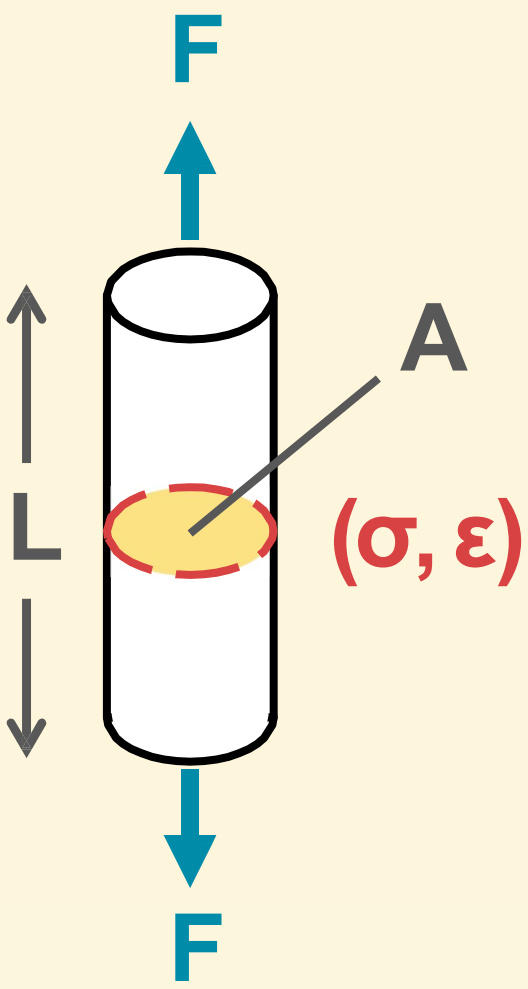


Illustration adapted from: Gordon, J.E., (1978), *Structures or Why Thing's Don't Fall Down*

Engineering stress and strain

- In uniaxial tensile testing, the applied force creates a stress that results in strain (elongation) of the material
  - Stress, $\sigma_n = \frac{\text{Force}}{\textbf{ORIGINAL} \text{ area}} = \frac{F}{A_0}$
  - Strain, $\varepsilon_n = \frac{\text{Change in length}}{\textbf{ORIGINAL} \text{ length}} = \frac{\Delta L}{L_0}$

- Engineering or **nominal** stress/strain calculated from the **original dimensions** of the part

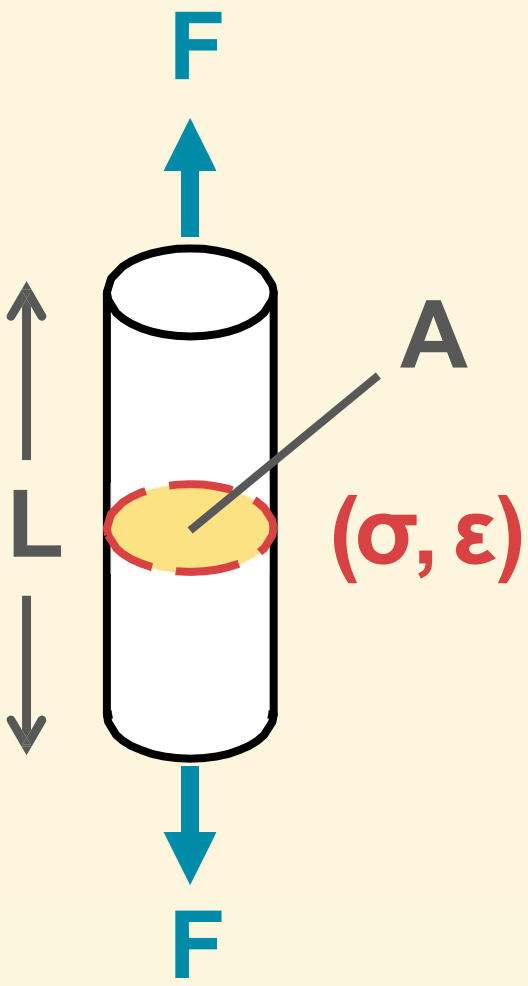


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True stress and strain (1)

- True tensile strain is given as:
  - Strain, $\varepsilon_t = \int_{L_0}^{L_i} \frac{dl}{l} = ln\frac{L_i}{L_0}$

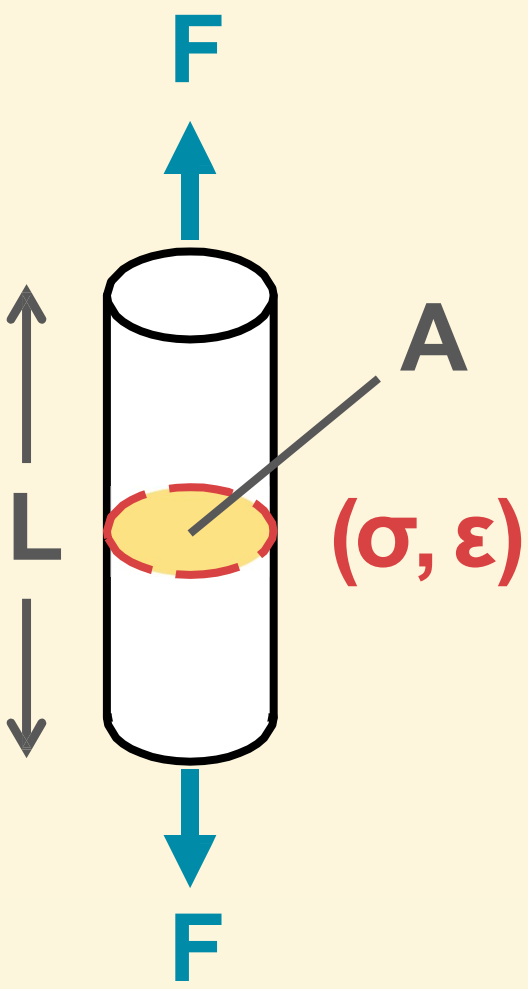
- As:
  - $\varepsilon_n = \frac{L_i - L_0}{L_0} \Rightarrow \frac{L_i}{L_0} = 1+\varepsilon_n$
  - $\boxed{\varepsilon_t = \ln(1+\varepsilon_n)}$



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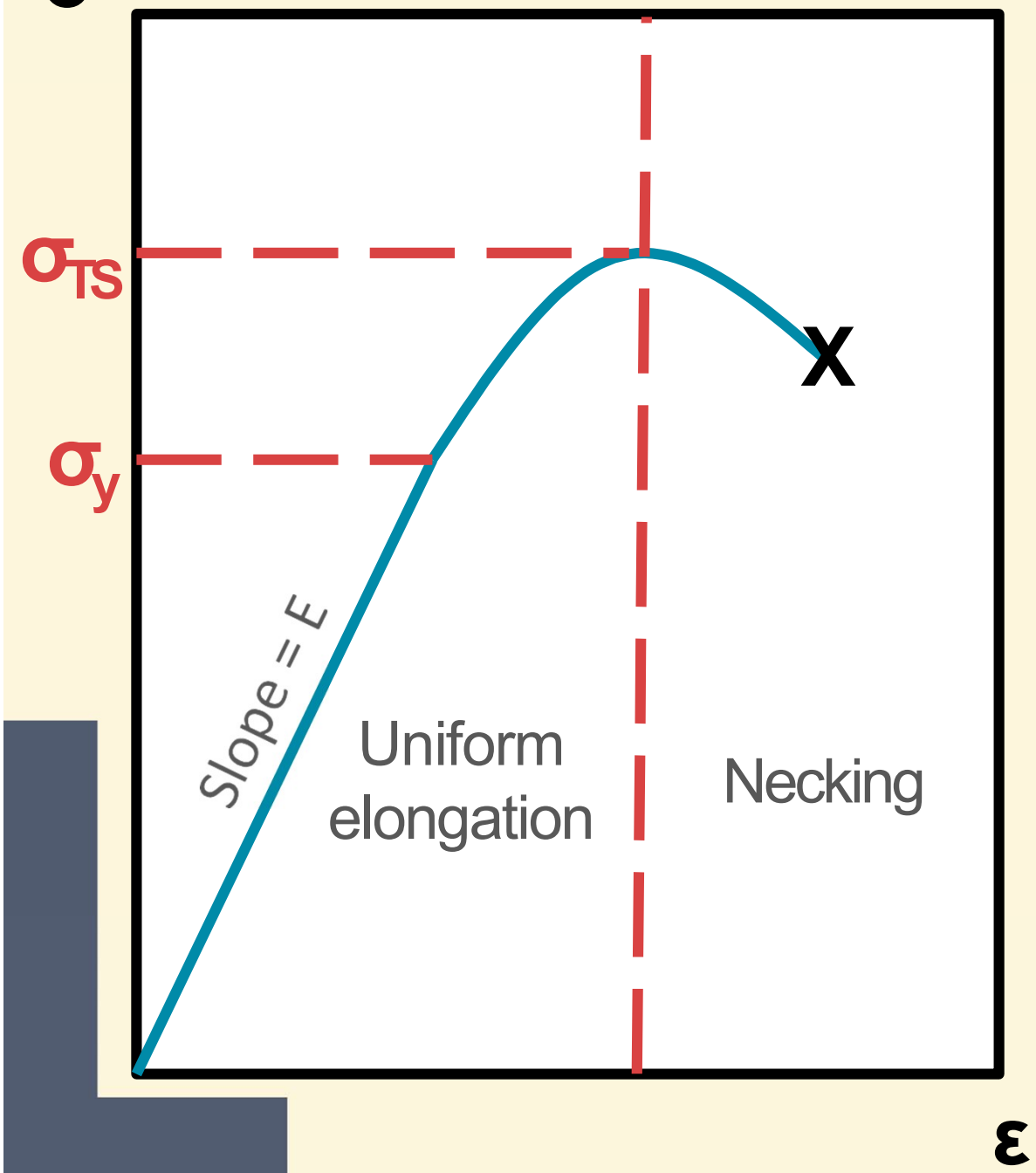
True stress and strain (2)

- True stress is calculated from the **instantaneous area** at any load:
  - Stress, $\sigma_t = \dfrac{\text{Force}}{\text{INSTANTANEOUS area}} = \dfrac{F}{A_i}$
- Neglecting volume change during plastic deformation:
  - $\dfrac{A_i}{A_0} = \dfrac{L_0}{L_i}$
  - $\sigma_t = \dfrac{F}{A_i} = \dfrac{F L_i}{L_0 A_0} \Rightarrow \sigma_t = \sigma_n \left(\dfrac{L_i}{L_0}\right)$
  - $\boxed{\sigma_t = \sigma_n(1 + \varepsilon_n)}$

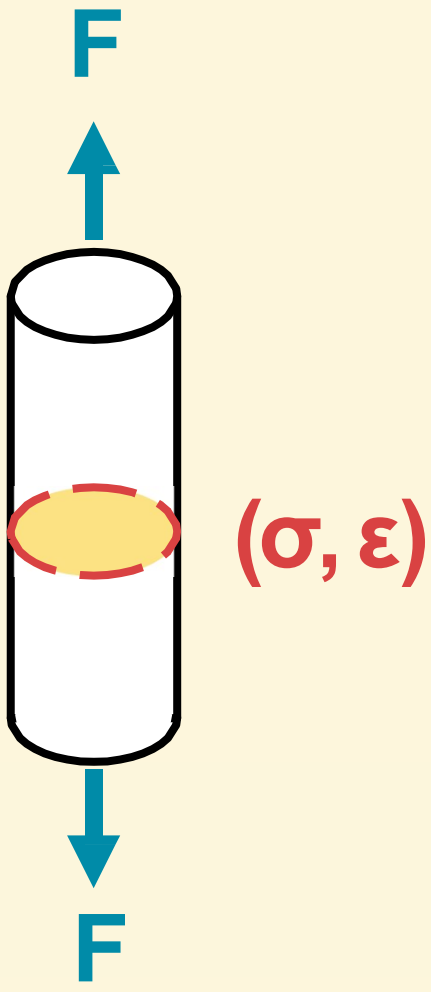


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Total strain, $\varepsilon = \varepsilon_e + \varepsilon_p = \frac{\sigma_y}{E} + \varepsilon_p$



Deformation of ductile materials



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Task!

- Search Young's modulus of various ceramics, metals, polymers and composites such as

  Ceramics: diamond, WC, Al_2O_3, SiC, Si_3N_4, MgO, ZrO_2, Silica, soda glass, alkali halides, cement, concrete, graphite, ice

  Metals: Osmium, tungsten, molybdenum, chromium, nickel, iron+steel, copper, titanium, aluminium, zinc, tin, magnesium, lead

  Polymers: Alkydes, melamines, PMMA, polystyrene, nylon, epoxy, polypropylene, rubbers, PVC, foamed polymers

  Composites: Cermets, CFRP's, fibreglass, GFRP's, woods

- Draw a bar chart data for Young's modulus of ceramics, metals, polymers and composites.